

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1127

Marble Tie 100 kV
April 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1127. The Interconnection Customer proposes a 114.8 MW solar photovoltaic facility interconnecting at the Marble Tie 100 kV substation in the Ohio area. The request was received on December 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 401 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$168,989,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Ripp - Idylwild 100.0 kV is overloaded to 383.9% of its 116 MVA emergency rating for the loss of the Beaver - Trego 500.0 kV. The project contributes 8.3% DFAX (9.6 MW) to this overload.

For the loss of Beaver - Trego 500.0 kV, the Axtel - Ripp 100.0 kV loads to 331.2% of its 142 MVA rating. Project DFAX: 8.3%.

The Rock Rapids - Idylwild 100.0 kV is overloaded to 324.5% of its 129 MVA emergency rating for the loss of the Beaver - Trego 500.0 kV. The project contributes 8.3% DFAX (9.6 MW) to this overload.

E Franklin - Whiskeytown 100.0 kV Line: 290.1% loading (174 MVA) following the outage of the Beaver - Trego 500.0 kV L; DFAX = 8.3%.

For the loss of Trego - Beaver 500.0 kV L, the E Franklin - Shinrock 100.0 kV loads to 281.4% of its 121 MVA rating. Project DFAX: 10.6%.

For the loss of Beaver - Trego 500.0 kV, the Smith Mtn - Havilah 100.0 kV line loads to 238.8% of its 100 MVA rating. Project DFAX: 7.0%.

For the loss of Trego Substation - Beaver 500.0 kV line, the Barre City - Nortech Substation 100.0 kV loads to 231.8% of its 123 MVA rating. Project DFAX: 23.6%.

Knight - Nortech 100.0 kV: 215.8% loading (147 MVA) following the outage of the Trego - Beaver 500.0 kV; DFAX = 7.0%.

For the loss of Trego - Beaver Substation 500.0 kV L, the Gren Hill - Smith Mtn 100.0 kV loads to 190.5% of its 116 MVA rating. Project DFAX: 7.0%.

Knight - Bayside 100.0 kV line: 185.5% loading (166 MVA) following the outage of the Beaver - Trego 500.0 kV; DFAX = 7.0%.

Heritage - Green Hill 100.0 kV: 163.2% loading (126 MVA) following the outage of the Trego - Beaver 500.0 kV; DFAX = 7.0%.

The Bayside - Emory Rd 100.0 kV is overloaded to 158.5% of its 120 MVA emergency rating for the loss of the N Campus - Bayside 100.0 kV. The project contributes 6.1% DFAX (7.0 MW) to this overload.

Emory Road - Oberlin Oh 100.0 kV Line: 156.1% loading (124 MVA) following the outage of the Bayside - N Campus Unm 100.0 kV Line; DFAX = 6.1%.

The Oberlin Oh - Velco Highgate 100.0 kV Line is overloaded to 155.4% of its 118 MVA emergency rating for the loss of the North Campus Unm - Bayside 100.0 kV Line. The project contributes 6.1% DFAX (7.0 MW) to this overload.

Havilah - Velco Highgate 100.0 kV: 155.4% loading (114 MVA) following the outage of the N Campus Unm - Bayside Substation 100.0 kV; DFAX = 6.1%.

N Campus Unm Substation - Bayside 100.0 kV: 154.7% loading (129 MVA) following the outage of the Oberlin Oh - Emory Rd 100.0 kV L; DFAX = 6.2%.

The Black Ranch - Chilcoat 100.0 kV L is overloaded to 151.7% of its 128 MVA emergency rating for the loss of the Trego - Beaver 500.0 kV. The project contributes 5.7% DFAX (6.5 MW) to this overload.

For the loss of Beaver - Trego 500.0 kV, the Bernis Substation - Black Ranch 100.0 kV loads to 150.8% of its 133 MVA rating. Project DFAX: 5.7%.

For the loss of Oberlin Oh - Emory Rd 100.0 kV line, the W Sta Nj - Havilah 100.0 kV Line loads to 150.2% of its 111 MVA rating. Project DFAX: 6.2%.

For the loss of Emory Rd - Oberlin Oh 100.0 kV, the West Station Nj - North Campus Unm 100.0 kV line loads to 148.5% of its 120 MVA rating. Project DFAX: 6.2%.

The Shinrock - Chilcoat 100.0 kV L is overloaded to 147.6% of its 120 MVA emergency rating for the loss of the Trego - Beaver 500.0 kV line. The project contributes 5.7% DFAX (6.5 MW) to this overload.

For the loss of Bernis - Black Ranch 100.0 kV, the Powersville Rd - Energy Shelby County 100.0 kV loads to 144.4% of its 140 MVA rating. Project DFAX: 6.6%.

For the loss of Black Ranch - Bernis 100.0 kV, the Timbrline - Powersville Road 100.0 kV L loads to 141.3% of its 152 MVA rating. Project DFAX: 6.6%.

For the loss of Black Ranch - Bernis 100.0 kV, the Flat Rock - Energy Shelby County Substation 100.0 kV line loads to 138.9% of its 138 MVA rating. Project DFAX: 6.6%.

Rock Rapids Substation - Firstenergy Edgewater 100.0 kV line: 138.5% loading (164 MVA) following the outage of the Beaver - Trego 500.0 kV; DFAX = 5.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ripp - Idylwild 100.0 kV	Rebuild line with higher-capacity conductor	\$6,341,000	100% (\$6,341,000)
Ripp - Axtel 100.0 kV	Rebuild line with higher-capacity conductor	\$1,924,000	100% (\$1,924,000)
Rock Rapids - Idylwild 100.0 kV	Rebuild line with higher-capacity conductor	\$2,298,000	100% (\$2,298,000)
Whiskeytown - E Franklin 100.0 kV	Rebuild line with higher-capacity conductor	\$4,982,000	100% (\$4,982,000)
Shinrock - E Franklin 100.0 kV	Rebuild line with higher-capacity conductor	\$6,528,000	100% (\$6,528,000)
Smith Mountain - Havilah 100.0 kV line	Rebuild line with higher-capacity conductor	\$4,076,000	100% (\$4,076,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barrecity - Nortech 100.0 kV	Rebuild line with higher-capacity conductor	\$21,643,000	100% (\$21,643,000)
Nortech - Knight 100.0 kV	Rebuild line with higher-capacity conductor	\$24,608,000	100% (\$24,608,000)
Gren Hill - Smith Mtn 100.0 kV	Rebuild line with higher-capacity conductor	\$4,285,000	100% (\$4,285,000)
Bayside - Knight 100.0 kV	Rebuild line with higher-capacity conductor	\$4,707,000	100% (\$4,707,000)
Heritage - Green Hill 100.0 kV	Rebuild line with higher-capacity conductor	\$5,269,000	100% (\$5,269,000)
Bayside - Emory Rd 100.0 kV	Rebuild line with higher-capacity conductor	\$953,000	100% (\$953,000)
Oberlin Oh - Emoryrd 100.0 kV	Rebuild line with higher-capacity conductor	\$831,000	100% (\$831,000)
Oberlin Oh - Velco Highgate 100.0 kV Line	Rebuild line with higher-capacity conductor	\$5,175,000	100% (\$5,175,000)
Velco Highgate - Havilah 100.0 kV	Rebuild line with higher-capacity conductor	\$4,745,000	100% (\$4,745,000)
N Campus Unm Substation - Bayside 100.0 kV	Rebuild line with higher-capacity conductor	\$1,970,000	100% (\$1,970,000)
Chilcoot - Black Ranch 100.0 kV	Rebuild line with higher-capacity conductor	\$12,523,000	100% (\$12,523,000)
Bernis Substation - Black Ranch 100.0 kV	Rebuild line with higher-capacity conductor	\$5,138,000	100% (\$5,138,000)
Havilah - W Sta Nj 100.0 kV	Rebuild line with higher-capacity conductor	\$3,597,000	100% (\$3,597,000)
West Station Nj - North Campus Unm 100.0 kV line	Rebuild line with higher-capacity conductor	\$4,626,000	100% (\$4,626,000)
Shinrock - Chilcoot 100.0 kV L	Rebuild line with higher-capacity conductor	\$23,666,000	100% (\$23,666,000)
Powersville Road - Energy Shelby County 100.0 kV Line	Rebuild line with higher-capacity conductor	\$8,800,000	100% (\$8,800,000)
Timbrline - Powersville Road 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,798,000	100% (\$1,798,000)
Flat Rock - Energy Shelby County 100.0 kV	Rebuild line with higher-capacity conductor	\$4,923,000	100% (\$4,923,000)
Rock Rapids Substation - Firstenergy Edgewater 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,583,000	100% (\$3,583,000)
	Total allocated to project		\$168,989,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.